Maritime Cybersecurity Awareness Training Study Material

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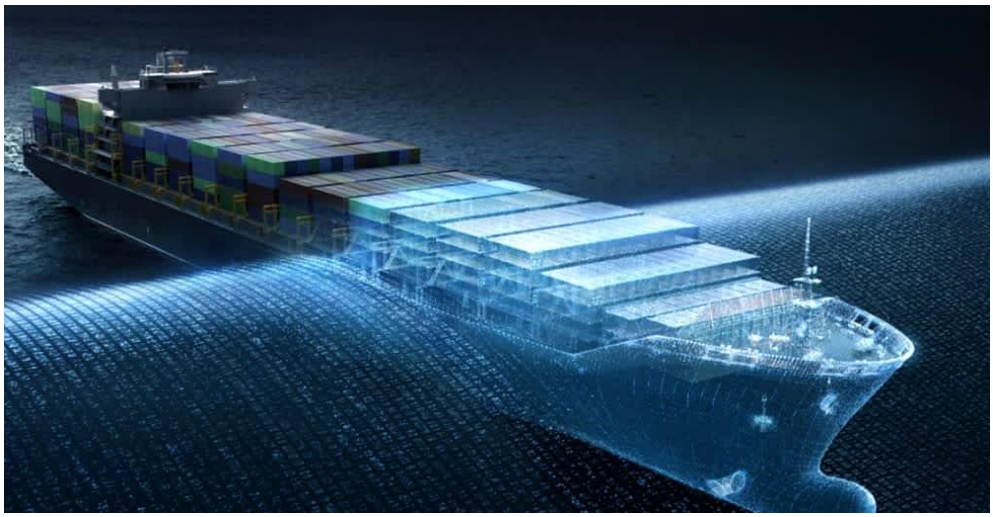
Maritime Cybersecurity Awareness Training

Executive Summary

Modern maritime vessels operate as **floating digital ecosystems**. While digitalization has enhanced operational efficiency and safety, it has simultaneously created new vulnerabilities. With over 90% of global trade transported by sea, ships have become attractive targets for cybercriminals and state-sponsored actors. This comprehensive guide addresses the critical need for maritime cybersecurity awareness among crew members and maritime professionals.



Modern Maritime Vessels Integrate Advanced Digital Systems for Operations

(Navigation, communication, engine control, cargo management, and safety systems are interconnected through digital networks.)

1. Introduction to Maritime Cybersecurity

1.1 Definition

Modern ships operate as floating digital environments where critical shipboard functions depend on interconnected digital technologies. Core operations such as navigation, engine and propulsion control, cargo handling, communication systems, and crew welfare services rely heavily on computers, onboard networks, and satellite-based connectivity. The increasing adoption of digital systems has significantly improved operational efficiency, safety, and decision-making capabilities. However, this same digitalization has also introduced new and often invisible cybersecurity vulnerabilities, exposing vessels to cyber threats that can originate from anywhere in the world.

1.2 The Reality of Cyber Risk at Sea

Unlike physical threats such as **fire, collision, or grounding**, cyberattacks may occur **without immediate physical warning signs** Cyber risks in the maritime domain differ from traditional maritime hazards. They are often **silent, unseen, and difficult to detect until damage has already occurred**.

Key realities include:

- Cyber threats in maritime environments are **invisible but highly dangerous**
- Over **90% of global trade is transported by sea**, making vessels attractive targets for cybercriminals and state-sponsored attackers
- A single cyber incident can:
    - Manipulate navigation systems
    - Disable engines or propulsion
    - Disrupt cargo operations
    - Interrupt global supply chains
- These risks highlight why cybersecurity must be treated as a **core component of maritime safety management**.

1.2.1 Key Truth for Crew

Cybersecurity is not only an IT responsibility. It is a critical safety issue, comparable in importance to fire prevention, emergency preparedness, and collision avoidance.

Simple human actions can unintentionally trigger serious cyber incidents, including:

- Clicking on malicious emails or links
- Using infected USB devices
- Creating weak or reused passwords
- Ignoring cybersecurity procedures

Such actions may lead to severe consequences, including:

- Loss of vessel control
- Environmental pollution and regulatory violations
- Significant financial and operational losses

- Direct threats to crew safety and human life

1.2.2 Prevention and Awareness

The majority of maritime cyber incidents are **preventable** by:

- Improving cybersecurity awareness
- Maintaining basic digital hygiene
- Strictly following company and shipboard cybersecurity procedures the risk of cyberattacks can be **significantly reduced**.

2. Why Cybersecurity Matters in Maritime

2.1 Navigation Safety: When the Ship Is Led Astray

Cybersecurity is no longer a technical or administrative concern—it is a **core maritime safety, environmental, and business risk**. Cyber incidents at sea can directly affect navigation, crew safety, environmental protection, regulatory compliance, and global trade continuity.

Modern vessels rely extensively on digital navigation systems such as **GPS, ECDIS, and AIS** to ensure safe passage. These systems are trusted by bridge teams for real-time positioning, route planning, collision avoidance, and situational awareness.

Cyber attackers can **manipulate or spoof navigation data**, causing vessels to unknowingly deviate from their intended route while all systems appear to function normally.

Potential Impact:

- Collisions with other vessels
- Grounding incidents
- Entry into restricted or high-risk waters
- Loss of situational awareness without visible warning signs

Modern ships rely on GPS, ECDIS, and AIS for safe passage. Cyber attackers can manipulate navigation data, causing vessels to sail toward danger while all equipment appears normal.

Figure 2: *Ship Bridge Navigation Systems – ECDIS and GPS Critical to Safe Operation*

Impact: Collisions, grounding, or entry into restricted waters without visible warning signs.

2.2 Crew Safety: When Systems Turn Against You

Shipboard operational technology (OT) systems such as **engine automation, ballast control, and power management systems** are increasingly automated and network-connected. If compromised, these systems can behave unpredictably or fail completely.

Cyber incidents may result in:

- Unexpected engine shutdowns
- Unstable or incorrect ballast operations
- Total loss of electrical power (blackout)

Such failures can escalate rapidly into **life-threatening emergencies**. In many cases, a single human error—such as clicking a malicious email or connecting an infected device—can put every life on board at risk.

2.3 Environmental Protection: When Nature Pays the Price

Cyberattacks can directly impact systems responsible for environmental safety and pollution prevention. A compromised system may trigger or fail to prevent:

- Oil spills
- Fuel leaks
- Release of hazardous or dangerous cargo

These incidents can cause **irreversible damage to marine ecosystems**, lead to severe legal consequences, and permanently harm the company's reputation.

Environmental protection is therefore inseparable from **cyber risk management**.

2.4 Business Continuity: When the World Stops Moving

A cyber incident does not only affect a single ship—it can disrupt **entire supply chains**.

Case Study: The NotPetya Cyberattack on Maersk (2017)

- Ports and terminals worldwide were shut down
- Operations were halted for several days
- Estimated financial losses of **USD 250–300 million**

- Significant disruption to global trade and logistics

This incident demonstrates that **even the largest and most technologically advanced organizations are vulnerable**. If a global shipping leader can be crippled, **any vessel or company is at risk**.

2.5 Regulatory Compliance: Cybersecurity Is Now the Law

Cybersecurity is a **mandatory regulatory requirement** in the maritime industry.

The **International Maritime Organization (IMO)** mandates cyber risk management under **IMO Resolution MSC.428(98)**. Shipping companies must integrate cyber risk management into their **Safety Management System (SMS)**.

Failure to comply may result in:

- Vessel detention
- Regulatory penalties
- Loss of certification
- Operational restrictions or bans

Cybersecurity compliance is therefore essential for **safe, legal, and uninterrupted operations**.

3. Cybersecurity Fundamentals

3.1 What Is Cybersecurity?

Cybersecurity means **protecting the ship's digital systems** from attackers who try to:

- Steal information
- Damage systems
- Take control of operations

A cyberattack does **not** need weapons or force. It can start with:

- A USB drive
- A malicious email
- An unlocked system
- Outdated software

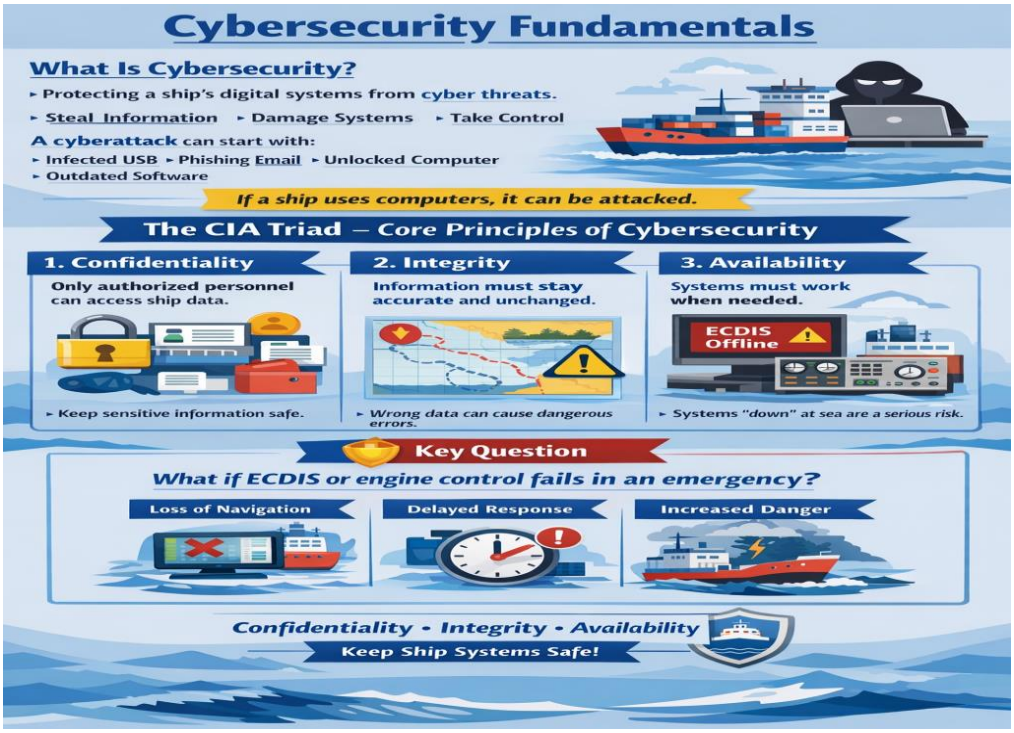
If the ship runs on computers, it can be attacked.

3.2 The CIA Triad: Three Principles for Digital Safety

Figure 3: The CIA Triad: Foundation of Digital Security

- **Confidentiality**: Only authorized people access ship data and systems
- **Integrity**: Information remains accurate and unaltered
- **Availability**: Systems work when needed, especially during critical operations

Ask yourself: What happens if ECDIS, engine control, or power systems are unavailable during an emergency?



3.3 Essential Cyber Terms for Seafarers

- **Malware**: Harmful software that damages or spies on systems
- **Ransomware**: Locks important files and demands money to unlock them
- **Phishing**: Fake messages designed to trick users into clicking links or sharing passwords
- **Firewall**: Digital security gate controlling what enters/exits ship systems
- **Patch**: Software update fixing known security holes
- **Vulnerability**: A weakness that attackers exploit, often caused by poor habits or outdated systems

Critical Insight: Attackers don't hack ships—they **exploit human mistakes**.

3.4 Three Pillars of Cyber Safety

Effective maritime cybersecurity is built on **three interconnected pillars**. These pillars work together to protect the vessel's digital and physical operations. If **any one pillar is weak or missing**, the entire ship becomes vulnerable to cyber threats.

Pillar	Description
People	Trained, alert crew members who understand cyber risks and think carefully before clicking links, opening emails, connecting USB devices, or sharing information.
Process	Clear and enforced procedures for safe use of computers, email systems, removable media, passwords, and incident reporting.
Technology	Properly configured shipboard systems supported by firewalls, antivirus software, access controls, network segregation, and regular system updates.

Table 1: *The Three Pillars – Interdependent Elements of Cybersecurity*

Why All Three Pillars Matter

- People alone are not enough without clear rules and technical protection
- Technology alone cannot prevent human error
- Processes fail if people do not follow them

Cybersecurity is effective only when People, Process, and Technology work together.

3.5 Maritime Safety Perspective

In the maritime environment:

- A well-trained crew (**People**)
- Following approved procedures (**Process**)
- Supported by secure and maintained systems (**Technology**)

creates a strong defence against cyber incidents—just like safety management for fire, collision, or pollution prevention.

If any one fails, the entire ship becomes vulnerable.

4. Common Cyber Threats at Sea

Figure 4: Cyber Threat Alert: Warning Signs of Digital Attacks

4.1 Phishing: The Most Dangerous Message You'll Ever Read

Attackers send **fake emails, SMS, or WhatsApp messages** pretending to be:

- Port authorities
- Company headquarters
- Charterers or suppliers

Red flags: Messages that create urgency or fear, demand immediate action, or request login details.

Defense: Verify sender through a secondary communication channel before acting.

4.2 Ransomware: Digital Hostage Taking

Ransomware locks important systems and files, demanding payment for restoration.

Real Maritime Incidents:

- **COSCO (2018):** Terminal operations disrupted, email systems locked
- **Port of San Diego (2018):** Critical IT systems compromised, port operations delayed

Impact: Imagine losing access to ship documents, cargo data, or operational systems mid-voyage.

4.3 Malware via Removable Media: The Infected USB Trap

USB drives, external hard disks, or contractor laptops can carry malware that:

- Infects navigation systems
- Spreads into OT networks
- Disables critical controls

A free USB can cost the ship millions.

4.4 GPS Spoofing: When the Bridge Is Lied To

Attackers transmit **fake GPS signals**, causing:

- Wrong position displayed on GPS/ECDIS
- False course and speed data
- Increased grounding and collision risk

The ship may be off-course while the screen shows "safe waters".

4.5 AIS Manipulation: Ships That Don't Exist

Attackers can:

- Hide real vessels
- Create phantom ships on AIS
- Broadcast false vessel identity or position

Impact: Poor collision avoidance decisions in congested waters.

4.6 Unauthorized Access: One Unlocked Screen Is Enough

Leaving systems unattended or protected with weak passwords allows:

- Physical system access
- Remote unauthorized access
- Data theft or manipulation

An unlocked console is an open door.

4.7 Social Engineering: When Attackers Sound Helpful

Attackers pose as:

- IT support staff
- Port inspection officers
- Company technicians

They use **politeness, authority, and urgency** to gain access.

If someone asks for passwords or system access, it's a red flag.

5. Maritime IT, OT & Integrated Systems

Modern vessels are **no longer separated into "office" and "engine room" worlds**—they are digitally connected ecosystems.

5.1 IT Systems: Where Most Attacks Begin

IT (Information Technology) systems include:

- Email and messaging systems
- Office computers and servers
- Crew Wi-Fi and internet access
- Business and documentation applications

These systems are **constantly exposed to the internet**, making them the **favorite entry point for attackers**.

Critical risk: An attack that begins in the "office" can travel silently toward critical ship systems.

5.2 OT Systems: Where Cyber Incidents Become Dangerous

OT (Operational Technology) systems control:

- Engine automation and propulsion
- Power management and generators
- Ballast and stability systems
- Cargo handling and monitoring

These systems were designed for **reliability, not cybersecurity**. Compromise can cause:

- Sudden engine shutdown
- Loss of propulsion or steering
- Power blackout
- Unsafe ballast conditions

When OT systems fail, there is no "reset button" at sea.

5.3 Integration Points: The Hidden Digital Bridge

Modern vessels connect IT and OT through:

- Satellite communication links
- Remote diagnostics and maintenance
- Shore-based monitoring and support

If access is not properly controlled: An attacker can move from email → server → OT system.

6. Critical Maritime Systems at Risk

6.1 ECDIS: When the Chart Lies

ECDIS (Electronic Chart Display and Information System) can be compromised through:

- Infected USB drives
- Unauthorized chart updates
- Unverified software patches

Always remember: If the data source is untrusted, the chart cannot be trusted.

6.2 GPS: Accurate Until It Isn't

GPS signals can be spoofed (fake signals) or jammed (blocked completely), resulting in:

- Wrong position, course, or speed
- False confidence during critical maneuvers

Golden Rule: Never trust GPS alone—cross-check with radar, depth sounder, and visual bearings.

6.3 AIS: Ships That Appear and Disappear

AIS data can be manipulated to:

- Create phantom vessels
- Hide real ships from view
- Broadcast false identity or location

If AIS traffic looks strange, verify with radar and VHF.

6.4 Engine Control Systems: No Second Chances

Engine control systems operate on OT networks and often lack modern cybersecurity protection. Compromise can cause:

- Sudden engine stoppage
- Loss of maneuverability
- Total blackout

These systems must be physically isolated and accessed only by authorized personnel.

6.5 Cargo Management Systems: Data Errors, Real Damage

Tampering can cause:

- Incorrect load distribution
- Stability issues
- Regulatory violations

Only authorized roles should access cargo data, and all changes must be logged.

7. Case Studies: Real Maritime Cyberattacks

7.1 Case Study 1: Maersk Not Petya Attack (2017)

Figure 5: Global Shipping Networks: Vulnerable to Coordinated Cyber Incidents

What Happened

A routine update of **trusted Ukrainian accounting software** carried hidden malware. Within minutes:

- 45,000 PCs and 4,000 servers destroyed
- 76 port terminals shut down worldwide
- Booking, cargo tracking, and documentation systems collapsed

Financial losses: $250-300 million

Key Lesson

The attacker did not break in—they were invited in through trusted software.

Network segmentation alone could not stop the attack, but **offline backups** saved the company. One domain controller in Ghana that was offline during the attack became the only source to rebuild the entire network.

Critical insight: If that power outage hadn't occurred, recovery could have taken months or failed entirely.

7.2 Case Study 2: COSCO Shipping Ransomware (2018)

What Happened: COSCO's North American network was hit by ransomware that locked systems and shut down email and booking platforms.

Impact: Staff forced to use manual, paper-based processes. However, strong regional network segmentation prevented spread beyond North America.

Lesson: Segmentation limits damage when incidents occur.

7.3 Case Study 3: Port of San Diego Ransomware (2018)

What Happened: Administrative systems including documentation and permits were compromised.

Impact: Vessel scheduling delayed, port operations backed up, ships waited longer at berth.

Lesson: Administrative systems are operationally critical. Offline and manual procedures must be available and tested.

8. Cyber Hygiene for Crew Members

Hygiene Practice	Crew Responsibility
Strong Passwords	Use unique, complex passwords and change periodically
USB Security	Never plug unknown drives into ship systems
Software Updates	Install only from authorized, verified vendors
Email Vigilance	Question messages creating urgency or fear
System Locking	Always lock computers when unattended
Early Reporting	Report unusual behavior immediately
Personal Devices	Avoid using public Wi-Fi on devices accessing ship systems

Table 2: Essential Cyber Hygiene Practices for Maritime Crew

8.1 Passwords: Your First Line of Defense

Use **strong, unique passwords** for ship systems. Never share them, even with colleagues.

If someone gets your password, they act with your authority.

8.2 USB Drives: The Silent Infection

Never plug **unknown or personal USB drives** into any ship system. Many cyber incidents begin with a seemingly "harmless" USB.

If you didn't bring it, approve it, or scan it—don't connect it.

8.3 Software Updates: Trust Only the Right Source

Install software and updates **only from authorized, verified vendors**. Fake or unofficial updates can contain hidden malware.

A wrong update can damage systems more than no update at all.

8.4 Emails & Messages: Urgency Is a Red Flag

Be cautious of emails or messages that:

- Demand immediate action
- Threaten consequences
- Ask for passwords or links

Always verify the sender through a second channel.

Workstations: Lock Before You Walk Away

Always lock computers when unattended. An unlocked system is an open door for misuse or attack.

If you leave it open, someone else controls it.

9. Securing Shipboard IT Systems

Digital security on IT systems requires multiple layers of protection:

9.1 Firewalls & Endpoint Protection

Firewalls act like the **hull of the ship**, keeping unwanted traffic out. If outdated, attackers walk in unnoticed.

An unprotected system is an open sea—anyone can enter.

9.2 Patching & Software Control

Attackers exploit **known vulnerabilities**. Best practices:

- Apply patches as soon as approved
- Maintain an approved software list

- Remove unauthorized applications

9.3 Access Control

Use **role-based access** so crew access only what they need. Enable **multi-factor authentication (MFA)** where possible.

If everyone can access everything, attackers can too.

9.4 Network Separation

Critical systems must **never share networks** with:

- Crew Wi-Fi
- Personal devices
- Entertainment systems

Crew Wi-Fi is for comfort. Operational networks are for safety.

9.5 Backups: Your Last Line of Survival

Regular backups ensure systems can be restored after an incident. Backups are only useful if they:

- Are stored securely
- Are offline or protected
- Are **tested regularly**

10. Securing Operational Technology (OT)

Operational Technology controls engines, power, ballast, and propulsion. **There are no second chances at sea.**

10.1 Isolation: Keep OT Off the Internet

OT systems must be **physically and logically isolated** from IT systems and the internet:

- No browsing
- No email
- No crew Wi-Fi
- No shortcuts

If OT is online, attackers are already closer than you think.

10.2 Remote Access: Convenience Is a Risk

Vendor remote access must be:

- Absolutely essential
- Time-limited
- Closely monitored
- Disabled immediately after use

Permanent remote access is an open hatch—not a service.

10.3 Change & Maintenance Records

Every change to OT systems must be:

- Authorized
- Logged
- Reviewed

If you don't know what changed, you don't know what can fail.

10.4 Risk Assessments & Exercises

Regular **tabletop exercises** help crew understand failure scenarios and practice decision-making before actual incidents occur.

11. Physical Cybersecurity Risks Onboard

Not all cyberattacks come through the internet. Some walk onboard **wearing safety shoes and carrying a laptop**.

11.1 Access to Bridge & Engine Control

The bridge and engine control room contain systems that **directly control the vessel**.

- Restrict access to authorized personnel only
- Never allow unsupervised access
- Challenge unfamiliar faces politely but firmly

If someone can touch the system, they can control it.

11.2 Network Closets & Cables

Network cabinets, switches, and cable runs are often poorly secured and poorly labeled. An attacker can:

- Plug in a device
- Disconnect alarms
- Capture data silently

One small device in a cabinet can compromise the entire ship.

11.3 Technicians & Visitors

Before granting access:

- Verify **ID and port access badges**
- Confirm purpose with the Master or Chief Engineer
- Supervise at all times

Uniforms and confidence are not proof of authorization.

12. Cybersecurity During Port Calls

A vessel is **never more exposed** than during a port call. New people, new networks, time pressure, and routine work create **perfect opportunities for cyber intrusion**.

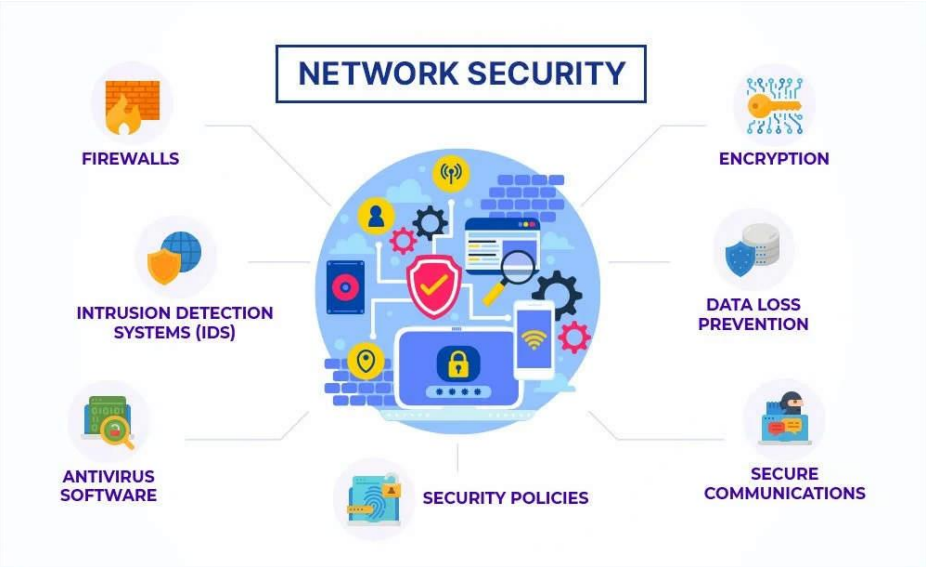


Figure 6: Network Security: Critical During Port Interfaces and External Connections

12.1 Port Networks: Convenience Can Be a Trap

Port and terminal networks are **outside your control**. Connecting ship systems to untrusted networks can allow:

- Malware entry
- Data interception
- Unauthorized access

If you wouldn't connect engine controls to public Wi-Fi, don't do it digitally.

12.2 Visiting Technicians: Supervise Every Move

Rules that must **never be broken**:

- Only **pre-approved devices** may be connected
- All devices must be **whitelisted and scanned**
- Activities must be **recorded and logged**
- No technician should work **unsupervised**

Unsupervised access is an open invitation to attackers.

12.3 Information Sharing: Less Is Safer

Share **only what is necessary**, nothing more:

- Avoid full system details
- Do not share passwords or network layouts
- Follow **data minimization principles**

The more information shared, the more attackers can exploit.

13. Cyber Incident Reporting & Response Procedures

Cyber incidents do not start with explosions or alarms—they start with **something that "doesn't look right."** Your response in the **first few minutes** decides whether the ship stays safe or loses control.

13.1 Incident Response Flowchart

Figure 7: Incident Response: Step-by-Step Guide to Managing Cyber Emergencies

Step 1: Identify & Contain

If you notice:

- Pop-ups
- System slowdown
- Unexpected reboots
- Strange messages or file locks

Immediately isolate the affected system from the network if it is safe to do so.

Step 2: Inform

Notify **immediately**:

- The Master
- The designated Cyber Officer
- Shore IT / Security team

Do NOT try to fix the issue alone.

Step 3: Preserve

- **Do NOT power off critical systems** unless instructed
- Avoid rebooting affected systems
- Preserve logs, alerts, and on-screen messages

Evidence helps experts stop the attack and prevent repeat incidents.

Step 4: Document

Record:

- Time of first symptom
- What you saw
- What actions were taken
- Any error messages

Your notes may be more valuable than the system itself.

Step 5: Communicate

If email or ship systems are down:

- Use phones, satellite calls, or other **out-of-band communication**
- Follow the approved communication plan

Step 6: Learn & Strengthen

Once systems are restored:

- Review what went wrong
- Update procedures
- Brief the crew
- Fix gaps before attackers return

14. IMO Cybersecurity Regulations & Compliance

Cybersecurity is no longer a "best practice"—**it is a regulatory requirement** enforced by the International Maritime Organization (IMO).

Failure to comply can lead to:

- Audit findings
- Detentions
- Loss of certification
- Operational delays

14.1 IMO Resolution MSC.428(98)

This resolution mandates that **cyber risk management must be included in the Safety Management System (SMS)**.

Cybersecurity is treated just like:

- Fire safety
- Collision prevention
- Pollution control

If it affects safety, it must be managed—including cyber risk.

What Auditors Look For

During audits, inspectors verify whether the company and vessel have:

1. **Cyber risk assessments** - Do you know what systems are at risk?
2. **Crew awareness and training** - Do crew know how attacks happen and what to do?
3. **Incident reporting and response procedures** - Is there a clear plan when something goes wrong?
4. **Access control and system protection measures** - Are systems protected and properly managed?

An SMS without cyber risk is considered incomplete.

14.2 Cybersecurity: Not Just Documentation

Having policies is **not enough**. Cybersecurity must be:

- **Understood by the crew**
- **Practiced onboard**
- **Improved continuously**

Auditors may ask:

- "What would you do if systems go down?"
- "Who do you report to?"
- "Have you practiced this scenario?"

15. Cybersecurity Do's & Don'ts for Vessels

CYBERSECURITY DO'S	
Practice	**Benefit**
Use strong passwords and MFA	Prevents unauthorized access
Keep systems patched	Closes known security holes
Segregate networks (IT/OT)	Limits cyber incident damage
Verify updates and sources	Prevents malware distribution
Report suspicious activity immediately	Enables rapid response

CYBERSECURITY DON'TS	
Risk	**Consequence**
Do not share credentials	Compromises access control
Do not connect personal devices to ship systems	Introduces outside threats
Do not use unknown USBs	Risk of malware infection
Do not click suspicious links	Enables phishing attacks
Do not allow unsupervised vendor access	Permits unauthorized system access

Table 3: Cybersecurity Best Practices and Common Mistakes

WHAT TO DO!

Reporting Cyber Incident

Refer to Cyber Respons e Plan in ISMS

For Cyber Incident inform CISO (Chief Information Security Officer)

Cybersecurity@sy

For all vessel IT issues and requirement

vessel.it@synergy group.sg

16. Cybersecurity Assessment Quiz for Crew

Question 1: The CIA Triad

Name the three parts of the CIA triad and explain why each matter on a vessel.

Expected answer: Confidentiality (only authorized access to data), Integrity (data remains accurate and unaltered), Availability (systems work when needed, especially at sea).

Question 2: Ransomware Signs

List three signs of a ransomware incident on a ship or at a port.

Expected answer: Files locked with unusual extensions, ransom messages appearing, system slowdown, inability to access critical files or systems, unusual error messages.

Question 3: GPS Spoofing Response

What immediate steps should you take if you suspect GPS spoofing?

Expected answer: Stop relying on GPS alone, cross-check position with radar and visual bearings, notify the Master, verify course and position through multiple methods, report to shore if necessary.

Question 4: Network Segmentation

Why is network segmentation important? Give a maritime example.

Expected answer: Segmentation isolates critical systems (OT) from general networks, preventing cyber incidents from spreading. Example: If crew Wi-Fi is compromised, isolated OT networks remain protected.

Question 5: USB & Contractor Devices

What are safe practices for handling USB drives and contractor laptops?

Expected answer: Never connect unknown USB drives, scan all external devices before use, use whitelisted/approved contractor equipment only, supervise all external device connections, maintain logs of all connections.

Scenario: Social Engineering Attempt

Situation: An individual claiming to be "IT support" asks for your password to fix ECDIS.

Correct Response: Do not provide the password. Verify the person's identity through the Master or designated Cyber Officer. Follow company procedures for IT support. Never share credentials with anyone, regardless of authority or urgency.

17. Conclusion & Best Practices Summary

17.1 Cybersecurity Is Seamanship in the Digital Age

The modern ship sails on **steel, fuel, and data**. When data is compromised, **control is compromised**.

Cybersecurity is no longer a technical issue left to shore offices or IT teams. It is a **core part of maritime safety, business continuity, and legal compliance**.

17.2 Impact of a Single Cyber Incident

A cyber incident can:

- Mislead navigation (spoofing, data tampering)
- Shut down engines (malware, ransomware)
- Delay ports and cargo (system failures)
- Harm the environment (accidental releases)
- Endanger lives (loss of control)

Most importantly—it often starts with something small and preventable.

17.3 The Resilience Formula

$$\text{Resilience} = \text{Awareness} + \text{Preparedness} + \text{Segmentation} + \text{Backups}$$

- **Awareness** stops mistakes
- **Preparedness** enables fast response
- **Segmentation** limits damage
- **Backups** enable recovery

Miss one, and the entire defense weakens.

17.4 What Every Crew Member Must Remember

1. **Practice cyber hygiene** every day (strong passwords, locked systems, no unknown USBs)
2. **Understand the threats** (phishing, ransomware, spoofing, social engineering)
3. **Protect IT and OT systems** (segmentation, controlled access, approved updates)
4. **Learn from real incidents** (Maersk, COSCO, ports, GPS spoofing)

Technology fails. Awareness saves.

17.5 Final Reality Check

Cyber threats are:

- **Invisible** - No alarms or visible warning signs

- **Patient** - Attackers wait for the right moment
- **Always evolving** - New threats emerge constantly

They don't attack computers—they **attack trust, routine, and complacency**.

Final Message for Maritime Professionals

Just as you:

- Keep a proper lookout
- Follow safety procedures
- Prepare for emergencies

You must **protect the ship digitally**.

A cyber-aware crew is a safe crew.
A prepared ship is a resilient ship.

At sea, **cybersecurity is seamanship.**